a quarter of an hour during which things occur that are neither in our past nor in our future.

They are elsewhere. We had never before been aware of this ‘elsewhere’ because, next to us, this ‘elsewhere’ is too brief; we are not quick enough to notice it. But it exists, and it is real.

This is why it is impossible to hold a smooth conversation between here and Mars. Say I am on Mars and you are here. I ask you a question and you reply as soon as you’ve heard what I said; your reply reaches me a quarter of an hour after I posed the question. This quarter of an hour is time that is neither past nor future to the moment in which you’ve replied to me. The key fact about nature that Einstein understood is that this quarter of an hour is inevitable: there is no way of reducing it. It is woven into the texture of the events of space and of time: we cannot abbreviate it, any more than we can send a letter to the past.

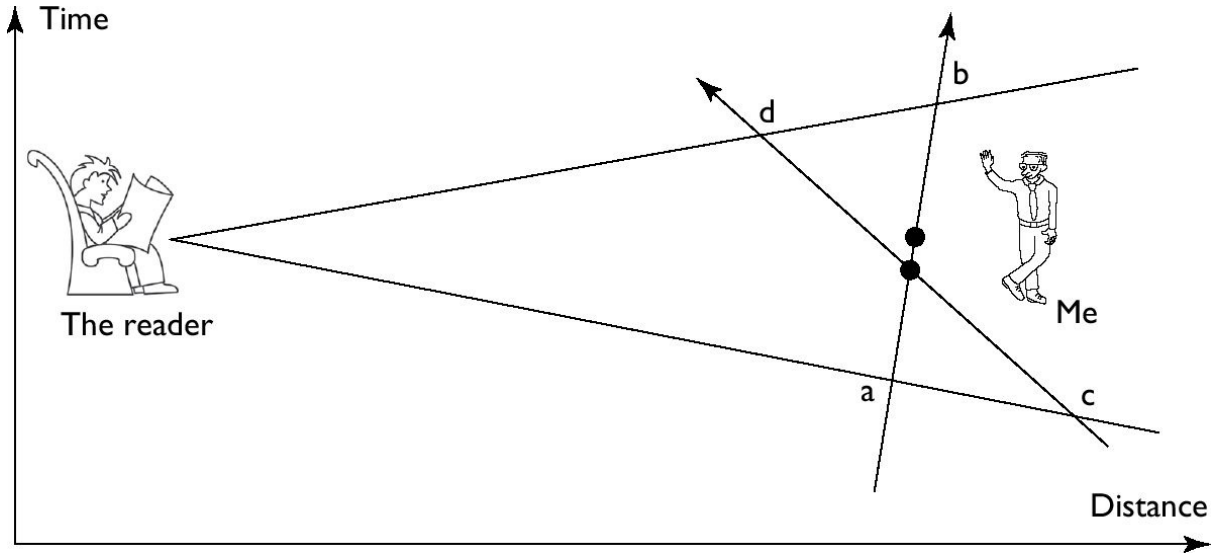


Figure 3.3 The relativity of simultaneity.

It’s strange, but this is how the world happens to be. As strange as the fact that in Sydney people live upside down: strange, but true. One gets accustomed to the fact, which then becomes normal and reasonable. It is the structure of space and time that is made like this.

This implies that it makes no sense to say of an event on Mars that it is taking place ‘just now’, because ‘just now’ does not exist (figure 3.3).[fn11] In technical terms, we say that Einstein has understood that ‘absolute simultaneity’ does not exist: there is no collection of events in the universe which exist ‘now’. The collection of all the events in the universe cannot be described as a succession of ‘now’s, of presents, one following the other; it has a more complex structure, illustrated in figure 3.2. The figure describes